

Total No. of Questions : 8]

SEAT No. :

PD-4079

[Total No. of Pages : 2

[6402]-39

S.E. (Computer Engineering)

DIGITAL ELECTRONICS & LOGIC DESIGN

(2019 Pattern) (Semester - III) (210245)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 & Q.7 or Q.8.
- 2) Figures to the right side indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.

Q1) a) Compare asynchronous counter & synchronous counter. [6]

b) Explain how J-K Flip-Flop is converted into [6]

i) TFF

ii) DFF

c) Draw & Explain 4 bit ripple up counter. [6]

OR

Q2) a) State the different types of shift Registers & state its any two applications. [6]

b) Write a short note on Johnson counter. [6]

c) What is Mod counter? Design Mod-9 using IC7490. [6]

Q3) a) Draw a block diagram of PLA device & Explain. [6]

b) Implement the following boolean function using PAL. [6]

$$f_1 = \sum m (3,6,7,11,12,13)$$

$$f_2 = \sum m (0,1,2,4,8,9,12,13)$$

c) What is difference between PLA & PAL. [5]

OR

Q4) a) What is ASM chart? State & Explain basic component of ASM chart. [5]

b) Implement following boolean function using PLA [6]

$$f_1 = \sum m (0,2,4,7,8,10)$$

$$f_2 = \sum m (1,2,3,8,9,10,14,15)$$

c) Design & Implement BCD to Excess - 3 code converter using PAL. [6]

P.T.O.

- Q5) a)** What is logic family? Give the classification of logic family & also write important characteristics of CMOS. [6]
- b)** Draw & Explain the circuit diagram of CMOS inverter. [6]
- c)** Define the following terms & mention the standard value of TTL logic family. [6]
- i) Fan out
 - ii) Power dissipation
 - iii) Noise Immunity

OR

- Q6) a)** Compare TTL & CMOS logic family. [6]
- b)** Explain CMOS NAND Gate with neat diagram. [6]
- c)** What are the advantages of open collector output? Justify your answer with suitable circuit. [6]
- Q7) a)** What is microprocessor & list different applications of microprocessor. [6]
- b)** Explain the memory organization of microprocessor. [6]
- c)** What are the different types of buses used in microprocessor & Explain it. [5]

OR

- Q8) a)** What are the various functional units of microprocessor? Explain ALU in brief. [6]
- b)** Draw & Explain block diagram of microprocessor. [6]
- c)** Explain 4-bit multiplier circuit using ALU & shift registers. [5]

